

Problem Description

Jenish and Neha are excellent friends. After performing several queries, Neha challenges Jenish to determine the highest possible Rating of the provided array A . According to Neha, the highest occurrence of an element in an array is its rating.

Jenish is given M and Q , the Magical Numbers. Jenish may perform Addition or Subtraction with M at most Q times for each element in the supplied Array A .

Because Jenish is stumped and unable to discover a solution, assist him in determining the highest possible ratings for the given array after applying queries to each element.

Constraints :

- $1 \leq N \leq 1000000$
- $1 \leq M \leq 100$
- $1 \leq Q \leq 10$
- $1 \leq A_i \leq 1000000$

Input :

- First line of Input contains integer M
- Second line contains integer Q
- Third line contains integer N
- Fourth line contains N elements representing elements of array A

Output :

- Output the highest possible rating of array A after applying Queries.

Explanation for test case 1

Jenish can add 1 in 1st element and subtract 1 from 3rd element to get highest frequency of 2 , i.e. 3

Logical Test Cases

Test Case 1

INPUT (STDIN)

1
1
4
1 2 3 4

EXPECTED OUTPUT

3

Test Case 2

INPUT (STDIN)

1
1
5
11 12 31 41 51

EXPECTED OUTPUT

2

Mandatory Test Cases

Test Case 1

KEYWORD

for(i=0;i<N;i++)

Test Case 2

KEYWORD

hash[A[i]]++;

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

9

Test Case 2

TOKEN COUNT

300

Test Case 3

NLOC

48

Code Editor

c

1 #include <stdio.h>
2 int main()
3

Custom Input (stdin)

T1 T2